

Name



ID#



Math 104 Spring 2025 Final Exam

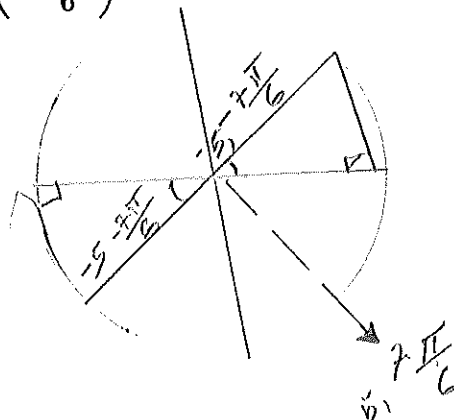
- A non graphing calculator is allowed
- Must show all work to receive credit
- Written or printed notes allowed (notes on digital devices not allowed)
- Attach all scratch paper

Question		Points
1	Polar, Vectors	/4
2	Average Rate of Change	/4
3	Trig Equations	/6
4	Side Side Angle Triangles	/6
5	Graphing, Domain, Range	/6
6	Applications	/6
Total		8 /30

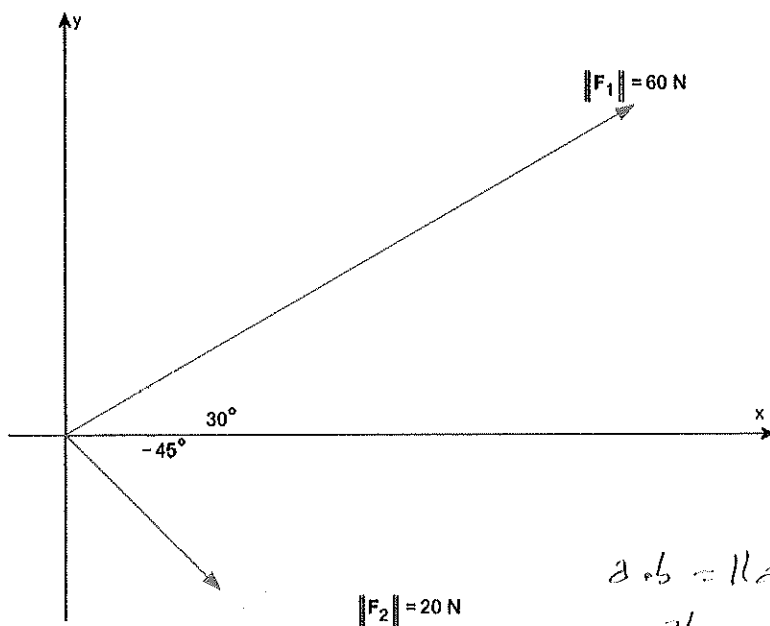
1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$

$R = 5$ for the distance

for $\theta > -\frac{7\pi}{6}$ $-\frac{7\pi}{6}$ 270°



1b) Add the two vectors $F_1 + F_2$



CROSS
V.K.W
 $\checkmark = 60 \hat{i} - 20 \hat{j}$

$$a \cdot b = \|a\| \|b\| \cos \theta$$

$$\frac{a \cdot b}{\|a\| \|b\|}$$

$$a \cdot b = 0 \|a\| \|b\|$$

$$60 \hat{i} + 20 \hat{j} ; 20 \hat{i} - 60 \hat{j}$$

$$\{60, 20\} \cdot \{20, -60\}$$

$$60 \cdot 20 + 20 \cdot (-60)$$

$$1200 - 1200 = 0$$

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2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3 .

$g(x) = 4 - x^2$ from $x = -1$ to $x = 3$ Also the slope of the secant line

$$\frac{g(3) - g(-1)}{3 - (-1)}$$

$$\frac{-5 - 3}{3 - (-1)} = \frac{-8}{4} = -2$$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

$$(-1, g(-1))$$

$$m = \frac{-5 - 3}{3 - (-1)} = \frac{-8}{4} = -2$$

$$y = -2x + 1$$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.

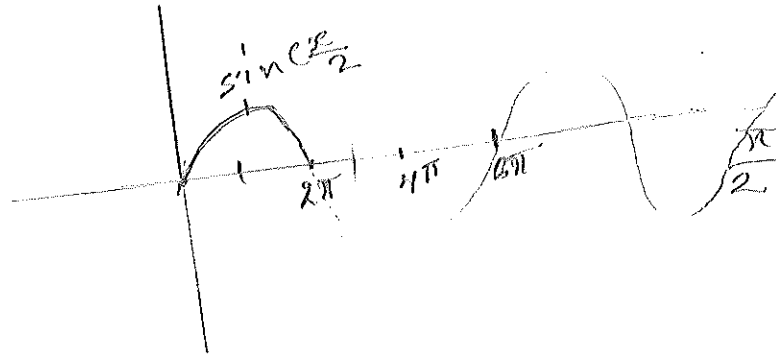
$$f(x) = -4\sin(x)$$

$$g(x) = 2\sin(x) + 3$$

$$f(x) \text{ max} = 0, \text{ min} = -4$$

$$g(x) \text{ max} = 2 + 3 = 5 \text{ min} = -2 + 3 = 1$$

$$T = \frac{2\pi}{1} = 2\pi$$



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$$f(x) = g(x) \text{ on the interval } [0, 2\pi]$$

$$f(x) = -4\sin(x)$$

$$g(x) = 2\sin(x) + 3$$

$$\sin(x) = -\frac{1}{2} \text{ in } [0, 2\pi]$$

$$x = \frac{7\pi}{6} \quad x = \frac{11\pi}{6}$$

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4) Two sides and an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$$b = 4, c = 5, B = 30^\circ$$

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$4^2 = 4^2 + 5^2 - 2 \cdot 4 \cdot 5$$

$$16 = 16 + 25 - 40 \cos A$$

$$16 = 41 - 40 \cos A$$

$$16 = 41 - 40 \cos A$$

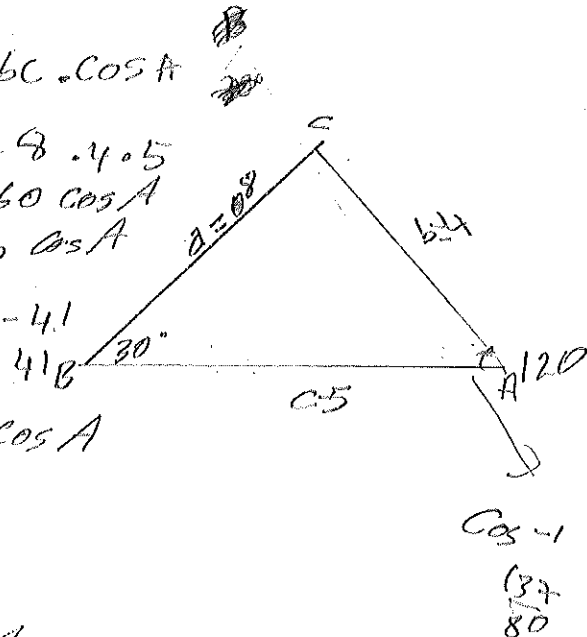
$$-41$$

$$-25 = -40 \cos A$$

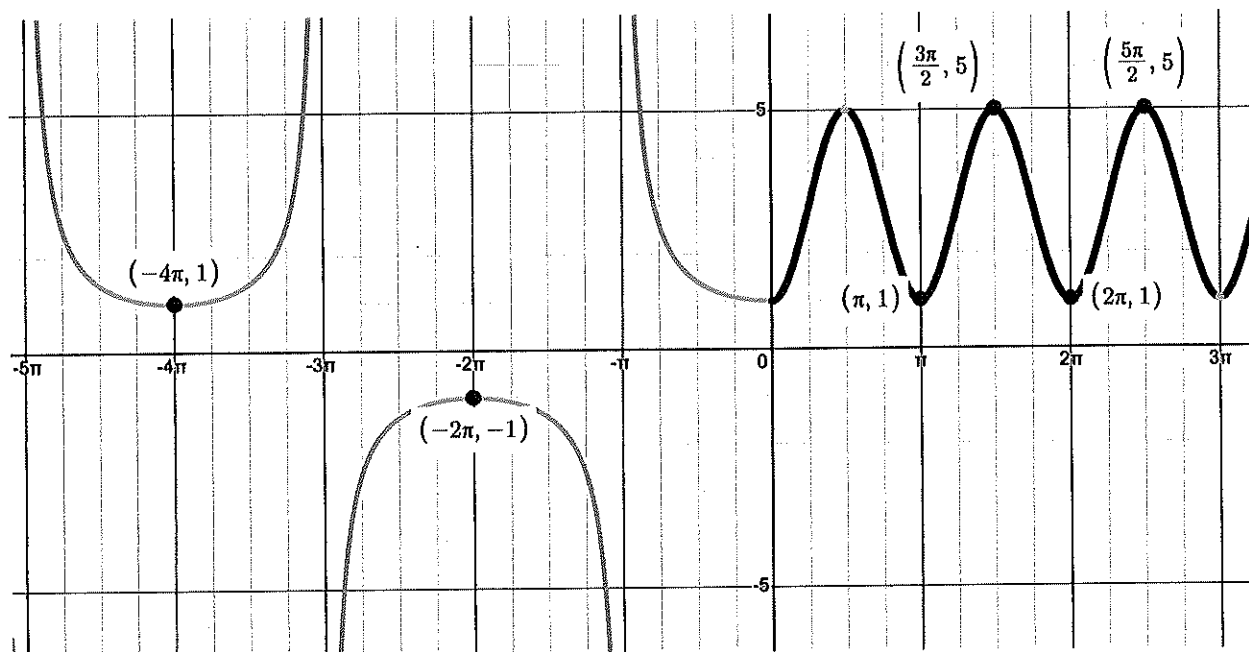
$$\frac{-25}{-40} = \cos A$$

$$\frac{25}{80} = \cos A$$

$$\cos^{-1} \left(\frac{25}{80} \right)$$



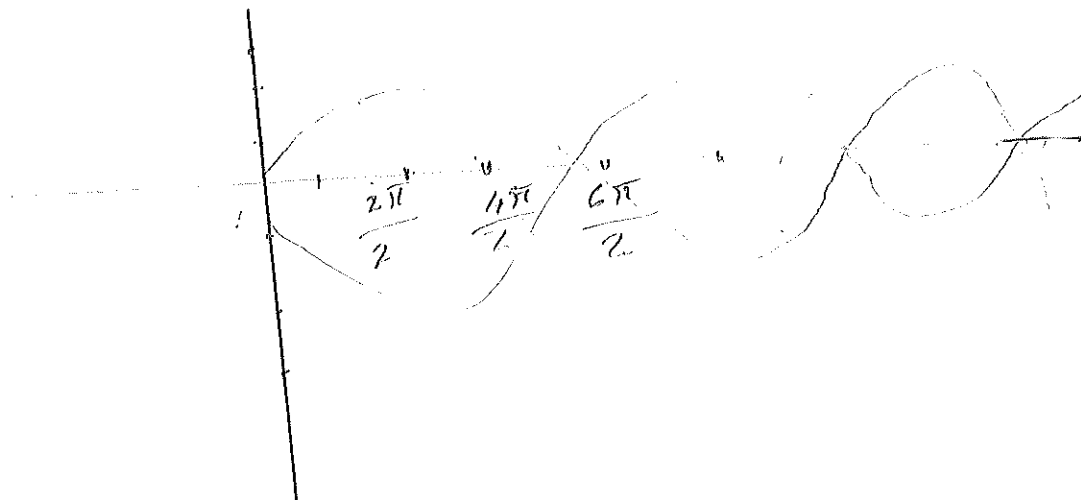
5) Given the graph below find the function $f(x)$ and state the domain and range.



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6) Blood pressure is modeled by the function $P(t) = 15 \sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.



b) The systolic pressure is 100

c) The diastolic pressure is 85

d) The heart rate is 15 beats per minute